Seat No.:_____

AHU ID No._____ Name_____ Major_____ Grade_____

Stony Brook Institute at Anhui University

2022 Spring Semester

Introduction to Object-Oriented Programming Midterm Exam

(Closed-book Exam Duration: 120 minutes)

Section	I	II	III	IV	V	VI	VII	Total score
Score								
Grader								

I. Multiple Choices (5 items, 2 points each) Score: ____

1. Assume x and y are declared properly as int variables and have been properly initialized. Which of the following are legal Java expressions? ( )
   A. boolean p = x > y > 0;
   B. x = y && y;
   C. x /= y;
   D. boolean p = (x != 0)&&(x == 0);

2. The allowed type of expression in a switch statement of Java is/are: ( )
   A. char B. short C. double D. String E. All the above

3. Which statement is true? ( )
   A. Pre-increment (++var) has higher precedence over post-increment (var++).
   B. Pre-decrement (--var) has the same level precedence as post-decrement (var--).
   C. Pre-increment (++var) has lower precedence over post-increment (var++).

4. Which of the following statements or expressions are incorrect? ( )
   A. boolean c = "Hello" >= "hi";
   B. String s3 = "Welcome to " + "Java";
   C. String s3 = "Welcome to Java" - "Java";
   D. boolean b = "Hello" != "hi";
   E. String s = "Welcome to Java";

5. You reserve memory locations for an array when you: ( )
   A. Declare the array name
   B. Use the keyword mem.
   C. Use the keyword new.
   D. Use the keyword size.

II. Programs analyzing (10 items, 4 points each) Score: ____

6. What will be the output of the following program?

```
public class Problem {
    public static void main(String[] args) {
        int x = Math.round(10/3.0);
        System.out.println(x);
        int y = 100/3;
        System.out.println(y);
         System.out.println(Math.rint(2.5));
        System.out.println(Math.rint(5.5));
    }
}
```

7. What will be the output for the following program:

```
public class Problem {
    public static void main(String[] args) {
        for(int i = 1; i < 15; i+=5) {
            System.out.print(i-- + " ");
      }
    }
}
```

8. What will be the output for the following program:

```
public class Problem {
    public static void main(String[] args) {
        boolean x = (3 + 4) * 3 < 5 * 4 + 3 - 5 || 12 - 3 * 4 > 5;
        System.out.println(x);
    }
}
```

9. What will be the output for the following program:

```
public class Problem {
   public static void main(String[] args) {
      int x = 20;
      float y = (x/100.0);
      double z = 50 * y;
      System.out.println(z);
   }
}
```

10. What will be the output for the following program.
[The ASCII value of '2' is 50]

```
public class Program {
     public static void main(String[] args) {
          System.out.println("1" + 1);
          System.out.println('2' + 1);
          System.out.println("3" + 4 + 5);
          System.out.println("4" + (5 + 6));
     }
}
```

11. Rewrite the following program using a switch statement.

```
public class Problem {
   public static void main(String[] args) {
       int x = 1, a = 3;
       if (a == 1)
           x += 5;
       else if (a == 2)
           x += 10;
       else if (a == 3)
           x += 16;
       else if (a == 4)
           x += 34;
       System.out.println("x:-->" + x);
      }
   }
```


12. What will be the output of the following program?

```
public class Problem {
     public static void main(String[] args) {
          int x = 4;
          int y = 5;
          int z;
          if(x++ == --y && ++y >= x) {
               System.out.println("inside if block !");
               System.out.println("x is :-->" + x);
               System.out.println("y is :-->" + y);
          }else if(++x == --y || y++ >= x) {
               System.out.println("inside else block !");
               System.out.println("x is :-->" + x);
               System.out.println("y is :-->" + y);
          }
          z = x++ + y;
          System.out.println("z is :-->" + z);
          }
}
```

13. What will be the output of the following program?

```
public class Problem {
      public static void main(String[] args) {
        double x = 2.99999;
        int y = (int)x;
        if(x = y)
            System.out.println("x is same as y!");
        else
            System.out.println(" x is different from y!");
      }
}
```

14. What will be the output of the following program?

```
public class Problem {
   public static void main(String[] args) {
        String st1 = new String("Stony Brook");
        String st2 = new String("Stony Brook University");
        System.out.println((st1 + " University" ) == st2);
   }
}
```

15. Write down a statement / statements to generate a random integer in the range from -55 to 15 (including -55 and 15). Do not write the complete program only the statement/statements for generating an integer in the required range.

III. Program Writing (10 points)

Score	

Write a program that simulates picking a card from a deck of 52 cards. Your program should display the rank (Ace, 2, 3, 4, 5, 6, 7, 8, 9, 10, Jack, Queen, King) and suit (Clubs, Diamonds, Hearts, Spades) of the card.

Here is a sample run of the program:

The card you picked is Jack of Hearts.

IV. Program Writing (10 points)

Score	

Write a program that accepts a string input from the user (through keyboard). Use this input string to create a new string that is composed of the characters present in the odd positions of the input string (in the order as it is present in the original input string). Here is a sample run:

Enter a string input: Beijing Chicago
Output string is: BiigCiao

[Hint: The odd position refers to the way we count in our day-to-day life, so the first character is counted as positioned at 1 and hence in odd position]

V. Program Writing (10 points)

Score	

Write a program that prompts the user to enter an integer for today's day of the week (Sunday is 0, Monday is 1, ..., and Saturday is 6). Also prompt the user to enter the number of days after today for a future day and display todays day (Sunday, Monday, ...) and the future day of the week. Here is a sample run:

Enter today's day: 1
Enter the number of days elapsed since today: 3
Today is Monday and the future day is Thursday

Enter today's day: 0
Enter the number of days elapsed since today: 31
Today is Sunday and the future day is Wednesday

VI. Program Writing (10 points)

Score	

Write a program that reads a commercial website URL from the keyboard; you should expect that the URL starts with www. and ends with .com. Retrieve the name of the site and output it. For instance, if the user inputs www.yahoo.com, you should output yahoo.

VII. Program Writing (10 points)

Score	

Suppose you save $100 each month into a saving account with the annual interest rate 5%. So, the monthly interest rate is p=0.05/12. After the n-th month, the value in the account becomes:

$S_n = (100+S_{n-1})*(1+p)$

where $S_0=0$. Write a program that prompts the user to enter an amount (e.g., 100), the annual interest rate (e.g., 5) and the number of months (e.g., 6) and displays the amount in the saving account after the given month.